

Yashasvi Ranjan

Senior Full Stack Software Engineer | Node.js | NestJS | React | TypeScript | GenAI / RAG | 9+ Years

+91 8553226608 | yashasvi1ranjan@gmail.com | Bengaluru, India | linkedin.com/in/yashasviranjan

SUMMARY

Senior Full Stack Software Engineer with 9+ years architecting scalable microservices, event-driven platforms, and production GenAI systems. Built GrowBot — a multi-source RAG chatbot with vector search and streaming LLM integration (Claude + Gemini) — deployed on Kubernetes and serving 3,000+ daily queries. Led backend service decomposition and reduced API latency by up to 97% through gRPC internal endpoints and Redis caching. Track record of presenting architecture proposals to Rakuten's Architect Guild, mentoring engineers, and owning end-to-end delivery for global platforms.

SKILLS

- Languages: JavaScript (ES2022+), TypeScript, Python, Node.js
- Backend: NestJS, Express.js, FastAPI, Flask, Loopback.js
- Frontend: React.js, Redux, Next.js, HTML5, CSS3, Webpack, Ant Design, HighCharts, Electron.js
- Databases: MongoDB, MySQL, PostgreSQL, Redis, SQLite, Couchbase, GridGain / Apache Ignite
- Messaging & Streaming: Kafka, RabbitMQ, BullMQ, WebSockets, Server-Sent Events (SSE)
- Auth & Security: Keycloak, OAuth 2.0, JWT, Passport.js, CASL (RBAC), Consul KMS, HashiCorp Vault
- Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes (CaaS), CI/CD, GitHub Actions, Jenkins, Terraform
- AI / GenAI: RAG Pipelines, ChromaDB, Vector Search (HNSW / cosine), Sentence Transformers, Claude API, Gemini API, Text-to-SQL, Prompt Engineering, SSE Streaming
- Practices: Microservices Architecture, Event-Driven Systems, System Design, Agile / Scrum, Technical Mentoring, Code Review

EXPERIENCE

Senior Software Engineer 2 — Rakuten India

Apr 2022 – Present

Lead developer — Grow LMS and internal Knowledge Base platform

- GrowBot (GenAI / RAG): Architected a production RAG chatbot serving 3,000+ daily queries using Python, FastAPI, ChromaDB, and Claude/Gemini APIs. Built multi-source ingestion (PDF, CSV, Excel, Confluence, live MySQL), role-based vector store access control with real-time permission resolution, SSE streaming responses, and schema-aware Text-to-SQL query routing. Deployed on Kubernetes (CaaS) with JWT and Keycloak authentication.
- Microservices Decomposition: Led the extraction of Knowledge Base, User Service, and a unified Notification Service (WebSocket + email) from a monolithic Flask/React codebase, resolving release-cycle conflicts and consolidating previously fragmented notification logic. Collaborated with senior engineers on parallel decomposition of Auth, Common, EL, and IL services — 9 microservices in total — supported by Consul (service discovery/config) and HashiCorp Vault (secrets management, key rotation) for securing sensitive HR data. Presented the platform's hybrid SaaS/IaaS multi-tenant architecture proposal to Rakuten's Architect Guild.
- API Latency Optimization (2–20s baseline to 50–300ms, up to 97% reduction): Diagnosed redundant inter-service JWT verification chains (3–4 network hops per request) and eliminated them via gRPC-based User Service endpoints and localized token verification. Implemented Redis caching for Keycloak access tokens (TTL-matched) and frequently-accessed user data. Presented the optimization at Rakuten's Asakai.
- Event-Driven Platform: Transformed Grow from a CRUD system into an event-driven platform — real-time WebSocket notifications, BullMQ queue-based processing, and scalable report generation pipelines for 10,000+ learners.
- RBAC Evolution: Implemented CASL-based RBAC for the Knowledge Base platform and evolved Grow's role model from binary Admin/User to a 6-persona system (Trainer, Admin, Super Admin, Report Analyst, Manager, User) with persona-specific dashboard views.
- Auth and Security: Built Keycloak SSO architecture across multiple repositories and deployments. Secured sensitive HR/psychometric data using Consul KMS and HashiCorp Vault with key rotation and encrypted S3 blob storage.

Software Engineer (Acting Team Lead) — Motorola Solutions

Apr 2020 – Mar 2022

Enterprise backend for Push-to-Talk devices — device communication, lifecycle management, and notification infrastructure

- Resolved 300+ open production bugs across KPNS (push notification) and LCMS (device lifecycle management) as sole contributor within 2 months of joining — recognized with a High Five and Bravo award.
- Given lead responsibility for a 4-engineer team spanning frontend, QA/automation, and backend integration; owned cross-team API contracts, code reviews, and release/deployment coordination across KPNS, LCMS, and UGWP.
- Migrated KPNS and LCMS from Loopback to Express with promise-based async patterns, extracting critical business logic (DB failover, health checks, notification triggers) that had been embedded inside node_modules — removing a significant maintainability and security-patching risk in an offline-build environment.
- Built rsync-based certificate synchronization between primary/secondary LCMS instances for PTT handset device authentication; diagnosed cross-instance data-consistency issues via network packet capture analysis.

Senior Software Engineer — Envestnet | Yodlee*Jul 2019 – Apr 2020*

- Designed an Investment Research Dashboard using HighCharts and Python Flask with advanced financial analytics and data visualization.
- Re-architected TXN application components, improving performance and maintainability.

Software Developer Analyst — IQVIA (IMS Health)*May 2018 – Jun 2019*

- Designed a microservice-based notification platform using Kafka, AWS SNS/SES, Docker, and Express.js.
- Built an e-consultation web platform using React.js, Express, and Twilio, and a cross-platform IoT health-tracking application.
- Built a cross-platform notification and registration desktop app using Electron.js in a 24-hour hackathon.

Software Engineer — Justdial*Jul 2016 – May 2018*

- Designed Mapzen Pelias data importers for Justdial's global map search using Node.js and Docker.
- Built a system to sync Instagram food-blogger content into Justdial's restaurant review ecosystem.
- Developed a SaaS-based POS desktop application for restaurants using Electron.js.

EDUCATION

Bachelor of Engineering — Information Science and Engineering

CMR Institute of Technology, Bangalore | 2012 – 2016